CUSTOMER SHOPPING BEHAVIOR ANALYSIS REPORT

Exploratory Data Analysis (EDA) & Business Insights

Item	Details
Project Type	Internship / Data Analytics Project
Dataset	customer_shopping_behavior(2).csv
Records	3,900 customer transactions
Variables	18 columns
Analysis Tools	Python, Pandas, Matplotlib, Microsoft Word
Prepared By	Manish Kumar Gautam
Organization / Internship	SAM AI Technologies
Date	5/10/2026

This report is prepared directly from the supplied customer shopping behavior dataset.

1. Executive Summary

This internship project analyzes 3,900 customer purchase records to understand customer demographics, purchasing patterns, product-category performance, seasonal behavior, payment preferences, shipping choices, subscription adoption, promotions, and purchase frequency. The dataset contains 18 fields and covers ages 18–70, purchase amounts from $20 to $100, and 50 locations.

Across all records, total recorded purchase value is $233,081, with an average transaction value of $59.76 and a median of $60. Clothing is the largest category by volume (1,737 purchases), while Spring has the highest seasonal transaction count (999). Male customers account for 2,652 records and female customers for 1,248. Payment methods are relatively balanced, with PayPal being the most used (677).

Key Findings at a Glance

- Customer age: mean 44.1 years; median 44 years.
- Average purchase amount: $59.76; total recorded purchase value: $233,081.
- Clothing leads all categories with 1,737 transactions (44.5% of all records).
- Spring has the highest transaction count at 999; Summer is lowest at 955.
- Male customers represent 68.0% of records; female customers represent 32.0%.
- PayPal is the most-used payment method (677), while Bank Transfer is the least-used (612).
- 1,053 customers have a subscription and 2,847 do not.
- 1,677 transactions show a discount applied; 1,677 show a promo code used.
- Average review rating is 3.75/5, indicating generally positive customer feedback.

2. Dataset Overview

The dataset represents customer shopping behavior at transaction level. Each row corresponds to a purchase record and includes customer demographics, item/category information, purchase value, location, season, review rating, subscription status, shipping, promotion usage, previous purchases, payment method, and purchase frequency.

Metric	Value
Rows / Transactions	3,900
Columns	18
Customer IDs	3,900 unique
Locations	50
Items Purchased	25 unique item names
Colors	25
Age Range	18–70 years
Purchase Range	$20–$100
Rating Range	2.5–5.0
Missing Values	0
Duplicate Rows	0

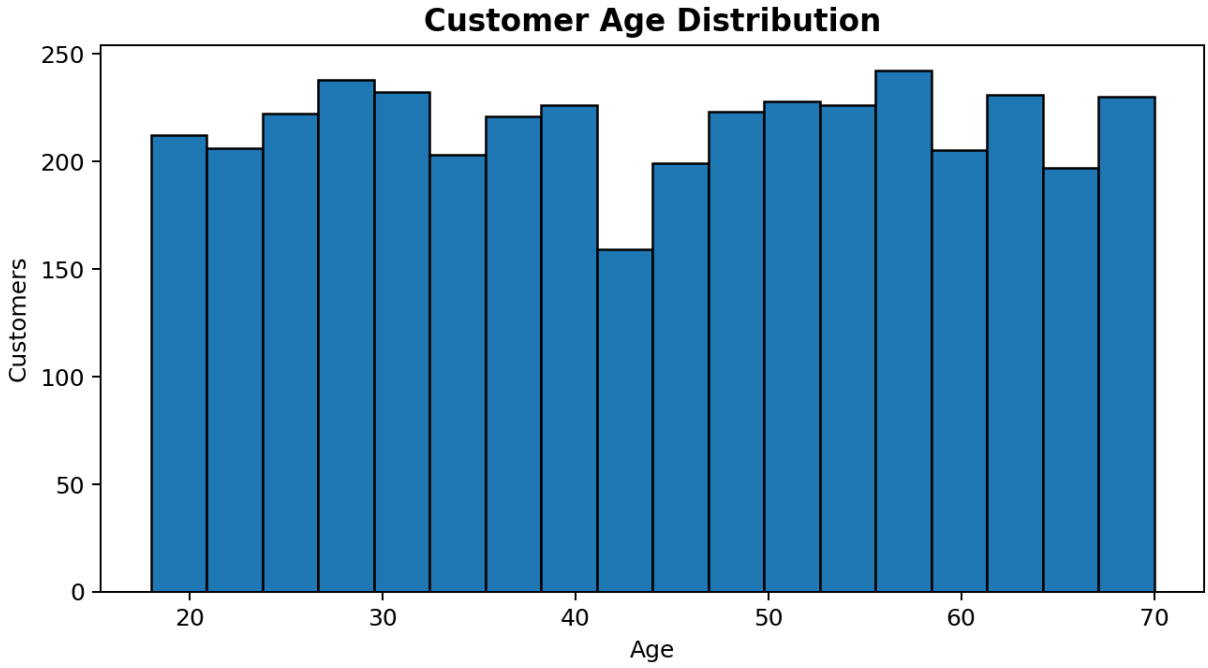
2.1 Data Quality Check

- No missing values were found across the 18 columns.
- No duplicate rows were found in the supplied dataset.
- Customer ID is unique for all 3,900 records.
- Data types are appropriate for analysis: numerical fields include age, purchase amount, review rating, and previous purchases; categorical fields cover customer and transaction attributes.
- No additional imputation was required for the supplied dataset.

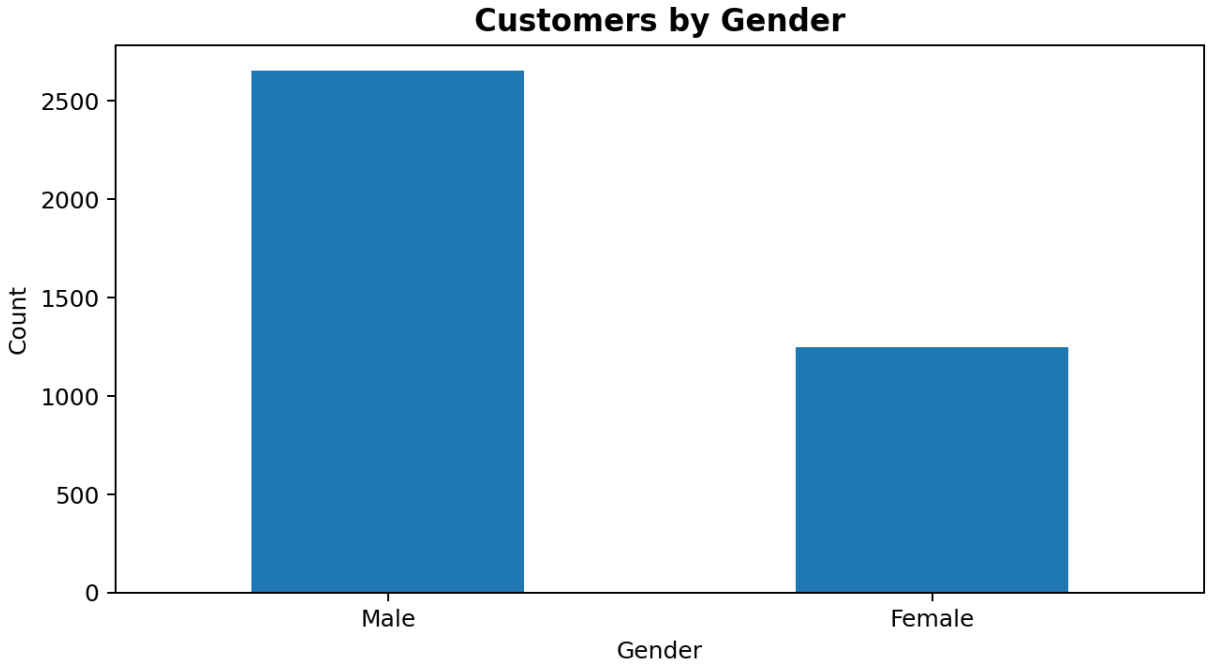
2.2 Data Dictionary

Column	Description	Type
Customer ID	Unique customer identifier	Integer
Age	Customer age	Integer
Gender	Customer gender category	Categorical
Item Purchased	Specific item purchased	Categorical
Category	Product category	Categorical
Purchase Amount (USD)	Transaction value in USD	Integer
Location	Customer location	Categorical
Size	Purchased item size	Categorical
Color	Purchased item color	Categorical
Season	Season associated with purchase	Categorical
Review Rating	Customer review rating	Float
Subscription Status	Whether customer is subscribed	Categorical
Shipping Type	Selected shipping method	Categorical
Discount Applied	Whether a discount was applied	Categorical
Promo Code Used	Whether a promo code was used	Categorical
Previous Purchases	Number of previous purchases	Integer
Payment Method	Payment channel used	Categorical
Frequency of Purchases	Customer purchase frequency	Categorical

3. Customer Demographics Analysis



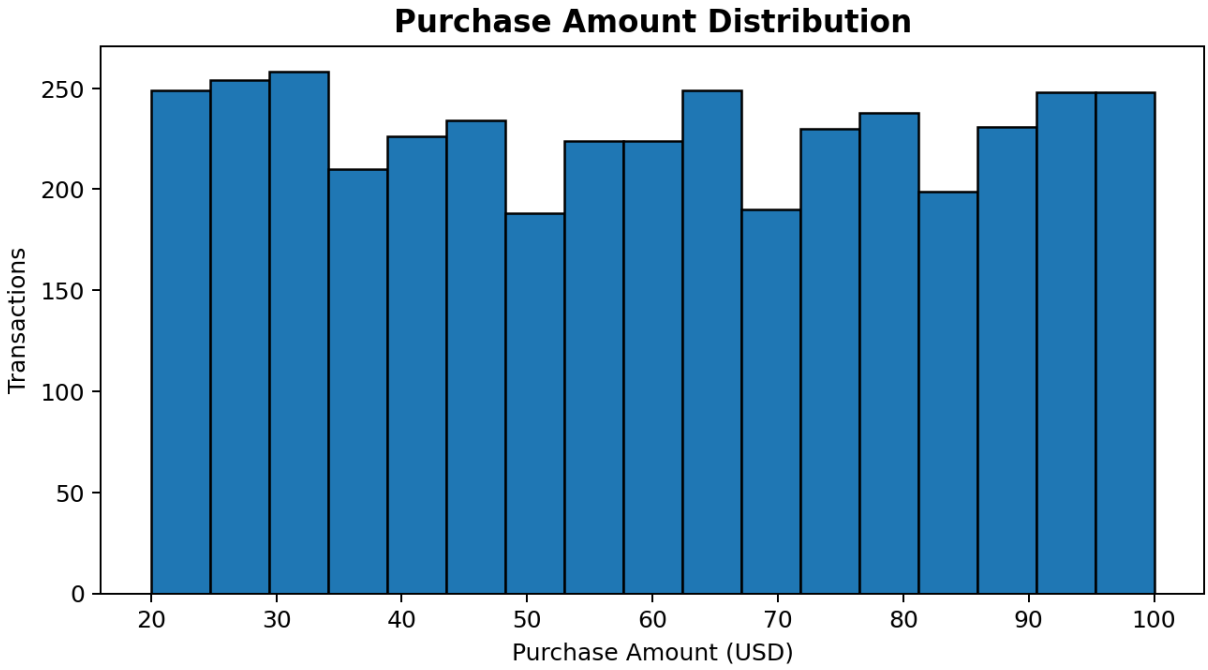
The customer age distribution spans 18 to 70 years. The average age is 44.1 years and the median is 44 years. The distribution is spread across the adult age range rather than being concentrated in only one narrow age group.



Gender-wise, the dataset contains 2,652 male records and 1,248 female records. Male records represent 68.0% of the dataset, while female records represent 32.0%.

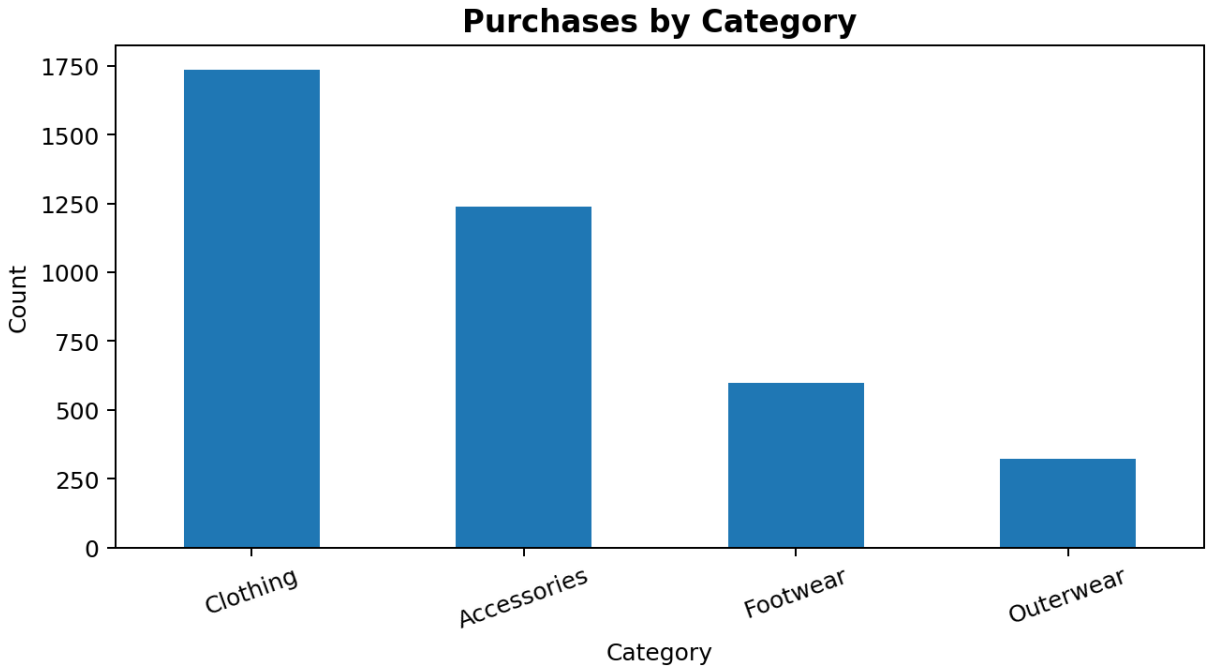
Gender	Transactions	Avg_Purchase	Total_Purchase	Avg_Rating
Female	1248	60.25	75191	3.74
Male	2652	59.54	157890	3.75

4. Purchase Behavior Analysis



Purchase amounts range from $20 to $100. The mean transaction value is $59.76, while the median is $60. The distribution is fairly broad and does not indicate a small number of extremely large purchases dominating the dataset.

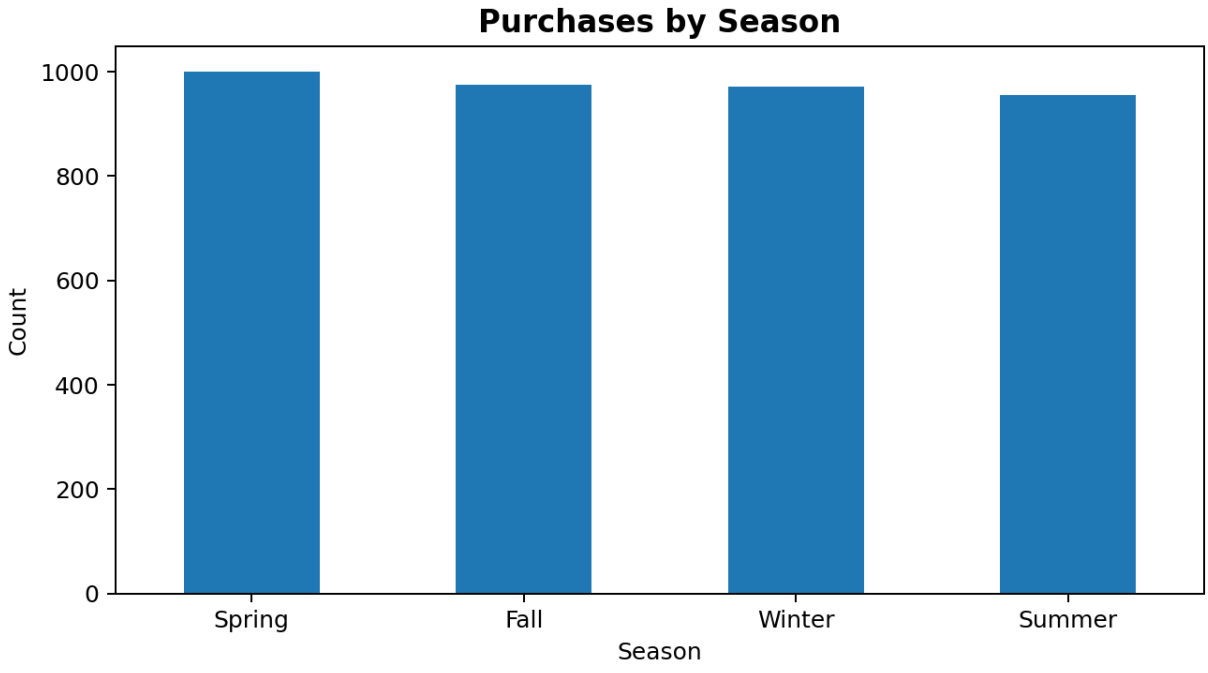
4.1 Category Performance



Category	Transactions	Share	Avg_Purchase	Total_Purchase	Avg_Rating
Clothing	1737	44.54	60.03	104264	3.72
Accessories	1240	31.79	59.84	74200	3.77
Footwear	599	15.36	60.26	36093	3.79
Outerwear	324	8.31	57.17	18524	3.74

Clothing is the strongest category by volume with 1,737 purchases (44.5%). Accessories follow with 1,240. Footwear contributes 599, while Outerwear has the lowest volume at 324. Footwear has the highest average purchase value among categories at $60.26.

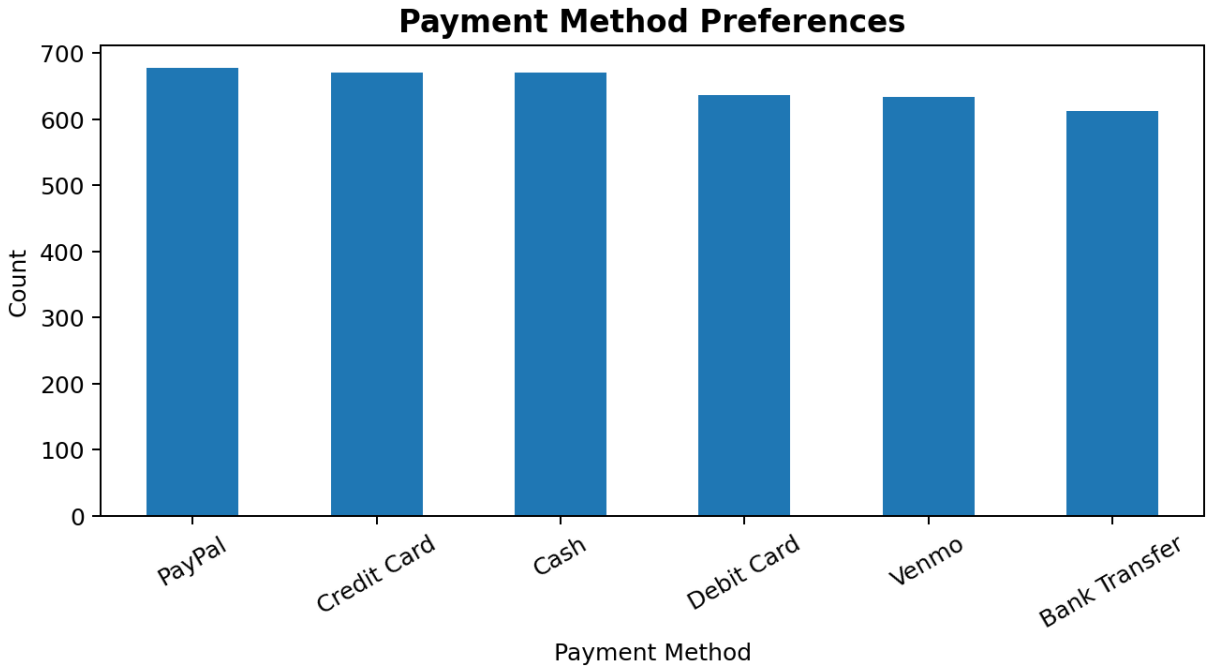
4.2 Seasonal Analysis



Season	Transactions	Avg_Purchase	Total_Purchase	Avg_Rating
Spring	999	58.74	58679	3.79
Summer	955	58.41	55777	3.72
Fall	975	61.56	60018	3.73
Winter	971	60.36	58607	3.75

Spring has the highest number of transactions (999), while Summer has the lowest (955). Fall records the highest average purchase value at $61.56. Seasonal transaction volumes are relatively close, suggesting that demand is distributed across the year.

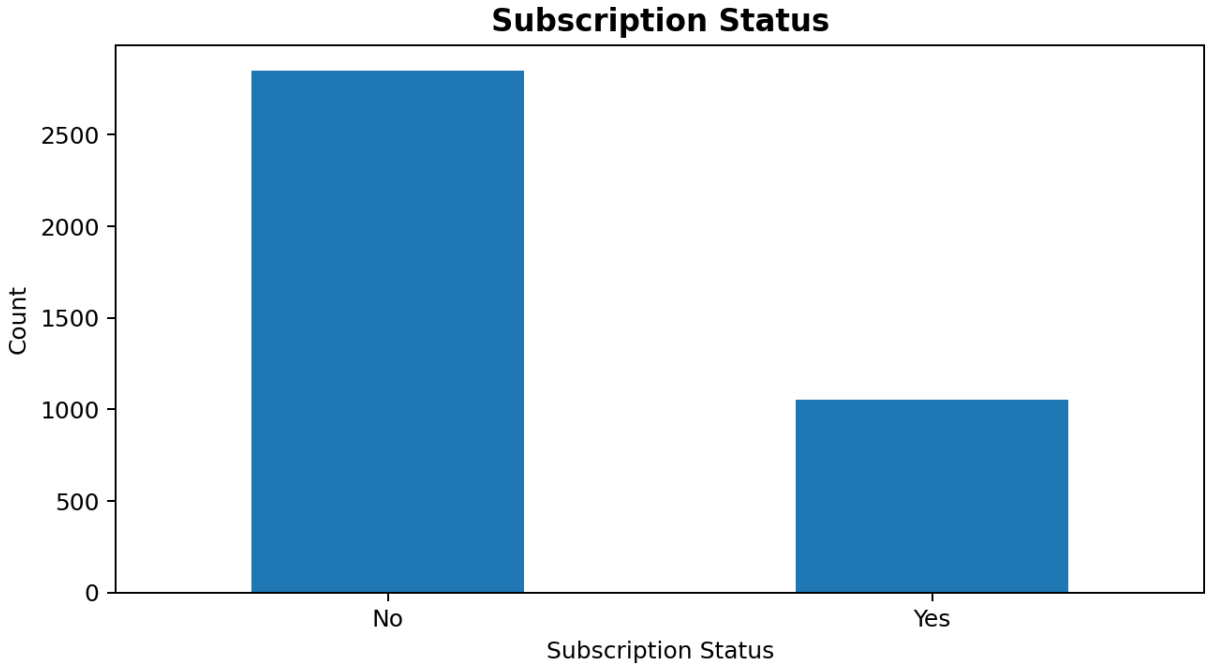
5. Payment Method Analysis



Payment Method	Transactions	Avg_Purchase	Total_Purchase
PayPal	677	59.25	40109
Credit Card	671	60.07	40310
Cash	670	59.7	40002
Debit Card	636	60.92	38742
Venmo	634	58.95	37374
Bank Transfer	612	59.71	36544

Payment usage is well distributed. PayPal is the most-used method with 677 transactions, followed by Credit Card (671) and Cash (670). Bank Transfer is the least-used with 612. Debit Card has the highest average transaction value among payment methods at $60.92.

6. Subscription, Discount & Promotion Analysis



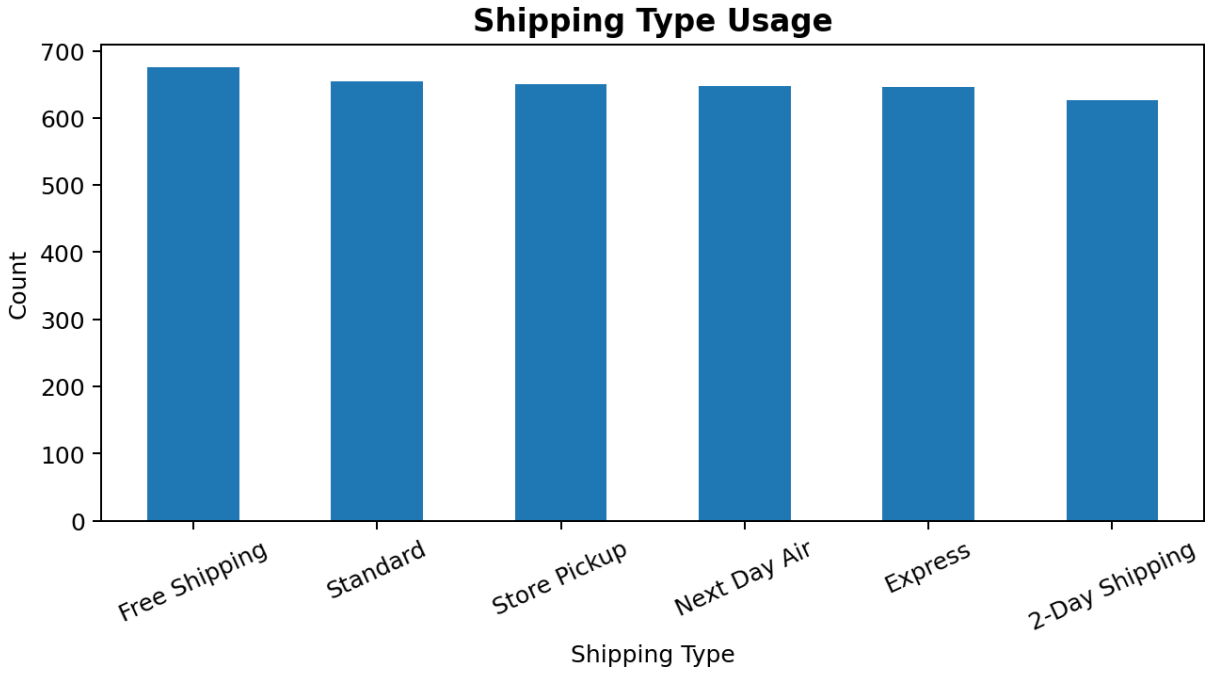
Subscription Status	Transactions	Share	Avg_Purchase	Total_Purchase
No	2847	73.0	59.87	170436
Yes	1053	27.0	59.49	62645

Only 1,053 records (27.0%) are marked as subscribed, compared with 2,847 non-subscribed records (73.0%). The average purchase values are similar between the two groups, so subscription status does not show a large transaction-value difference in this dataset.

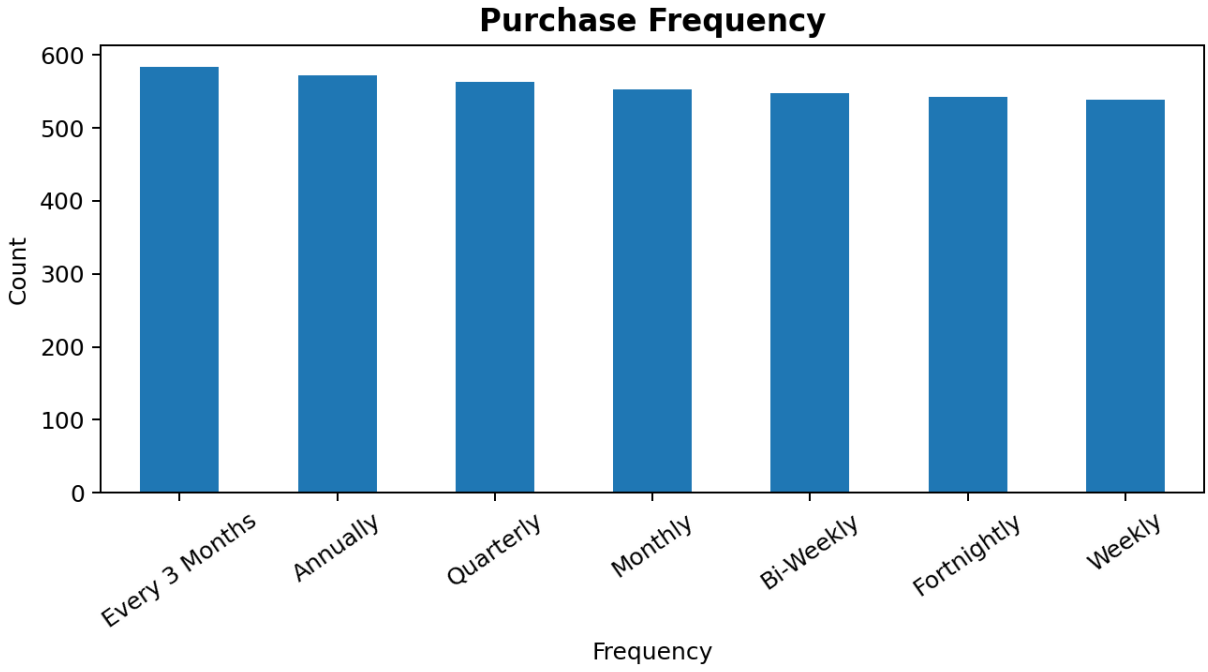
Promotion Indicator	Transactions	Share
Discount Applied = Yes	1,677	43.0%
Discount Applied = No	2,223	57.0%
Promo Code Used = Yes	1,677	43.0%
Promo Code Used = No	2,223	57.0%

Discount application and promo-code usage have identical counts in the dataset: 1,677 transactions use a discount/promo indicator and 2,223 do not. This suggests the two fields may be closely aligned in the source data.

7. Shipping & Purchase Frequency



Shipping Type	Transactions	Avg_Purchase	Avg_Rating
Free Shipping	675	60.41	3.71
Standard	654	58.46	3.82
Store Pickup	650	59.89	3.71
Next Day Air	648	58.63	3.72
Express	646	60.48	3.77
2-Day Shipping	627	60.73	3.76



Frequency of Purchases	Transactions	Avg_Purchase	Total_Purchase
Every 3 Months	584	60.08	35088
Annually	572	60.17	34419
Quarterly	563	59.98	33771
Monthly	553	59.33	32810
Bi-Weekly	547	60.69	33200
Fortnightly	542	59.05	32007
Weekly	539	58.97	31786

Purchase frequency is spread across weekly, fortnightly, bi-weekly, monthly, quarterly, every-three-months, and annual patterns. No single frequency dominates the dataset, indicating a mixed customer lifecycle.

8. Product, Location, Size & Review Insights

Top 10 items by transaction count:

Item Purchased	Transactions	Avg_Purchase	Total_Purchase
Blouse	171	60.88	10410
Pants	171	59.01	10090
Jewelry	171	58.54	10010
Shirt	169	61.14	10332
Dress	166	62.17	10320
Sweater	164	57.7	9462
Jacket	163	56.74	9249
Belt	161	59.84	9635
Coat	161	57.61	9275
Sunglasses	161	59.93	9649

Top 10 locations by transaction count:

Location	Transactions	Total_Purchase
Montana	96	5784
California	95	5605
Idaho	93	5587
Illinois	92	5617
Alabama	89	5261
Minnesota	88	4977
Nebraska	87	5172
Nevada	87	5514
New York	87	5257
Delaware	86	4758

Size distribution and performance:

Size	Transactions	Share	Avg_Purchase	Avg_Rating
M	1755	45.0	59.92	3.71
L	1053	27.0	58.56	3.75
S	663	17.0	61.04	3.79
XL	429	11.0	60.09	3.8

Average review rating is 3.75/5, with observed ratings ranging from 2.5 to 5.0. The average number of previous purchases is 25.35, which provides a useful indicator of customer repeat-purchase history.

9. Business Insights & Recommendations

Protect the Clothing category: Clothing generates the highest transaction volume and the highest category-level total purchase value. Maintain assortment depth, availability, and merchandising in this category.

Increase focus on Accessories and Footwear: Accessories is the second-largest category, while Footwear has a slightly higher average transaction value than the other categories. Cross-selling can combine these categories with Clothing.

Use seasonal campaigns: Transaction counts are relatively balanced across seasons. Rather than relying only on one peak season, campaigns can be planned continuously with seasonal product themes.

Maintain multiple payment options: Payment method usage is distributed across PayPal, Credit Card, Cash, Debit Card, Venmo, and Bank Transfer. Keeping multiple options reduces payment friction.

Improve subscription conversion: Only 27.0% of records are subscribed. Loyalty benefits, member-only offers, early access, or personalized recommendations could be tested to increase adoption.

Measure promotion effectiveness: Discount and promo-code indicators occur in the same number of records. Future analysis should test whether discounts increase conversion or repeat purchases while protecting average order value.

Optimize customer retention: Purchase frequency is distributed across several lifecycle patterns. Customer segmentation by frequency and previous purchases can support targeted retention campaigns.

Use location-level targeting: The dataset covers 50 locations. Location-specific demand analysis can be used for regional promotions, inventory planning, and shipping optimization.

10. Methodology / Project Workflow

- Data collection: imported the supplied CSV dataset.
- Data understanding: reviewed dimensions, columns, data types, and categorical values.
- Data quality: checked missing values, duplicate rows, and customer ID uniqueness.
- Descriptive analysis: calculated counts, percentages, averages, totals, ranges, and category-level summaries.
- Exploratory Data Analysis: examined age, purchase amount, category, season, gender, payment, subscription, shipping, frequency, product, and location patterns.
- Visualization: created histograms and bar charts to communicate major distributions and comparisons.
- Business interpretation: translated descriptive findings into practical recommendations for retail/customer analytics.
- Reporting: compiled the results into an internship-ready documentation report.

10.1 Tools & Technologies

Tool / Technology	Purpose
Python	Core analysis environment
Pandas	Data loading, cleaning, aggregation and analysis
Matplotlib	Data visualization
Microsoft Word / DOCX	Project documentation and presentation
CSV	Source dataset format

11. Limitations & Future Scope

- The dataset is transaction-level and does not provide a timestamp, so month-by-month or day-of-week trends cannot be analyzed.
- The data does not include profit, cost, margin, customer acquisition cost, or marketing spend, so profitability cannot be evaluated.

- The analysis is descriptive; it identifies patterns but does not establish causal relationships.
- Subscription and promotion fields should be investigated further because discount and promo-code counts are identical.
- Future work can include customer segmentation using RFM (Recency, Frequency, Monetary) analysis if transaction dates become available.
- Predictive models can be developed for purchase amount, repeat purchase probability, churn risk, or product recommendation.
- An interactive dashboard can be created in Power BI, Tableau, or Streamlit for management-level monitoring.

12. Conclusion

The customer shopping behavior dataset provides a useful view of 3,900 purchase records across demographics, products, categories, seasons, payment methods, promotions, shipping, and purchasing frequency. The analysis shows a broad customer age range, an average transaction value of $59.76, strong demand for Clothing and Accessories, balanced seasonal demand, and broadly distributed payment preferences.

From an internship/project perspective, the dataset demonstrates the complete analytics workflow: data inspection, quality validation, exploratory analysis, visualization, interpretation, and business recommendation. The resulting insights can serve as a foundation for customer segmentation, retention analysis, campaign optimization, and interactive business intelligence reporting.

Appendix A: Core Dataset Statistics

Statistic	Result
Total Transactions	3,900
Total Purchase Value	$233,081
Average Purchase	$59.76
Median Purchase	$60
Average Age	44.07
Average Review Rating	3.75 / 5
Average Previous Purchases	25.35
Most Common Category	Clothing (1,737)
Most Common Season	Spring (999)
Most Common Payment Method	PayPal (677)
Most Common Size	M (1,755)
Most Common Shipping Type	Free Shipping (675)
Most Common Frequency	Every 3 Months (584)

